Low-Complexity CNN Detection and Readhead Optimization in Dual-Reader Shingled Magnetic Recording Systems

Ekawee Anantawilailaka, Anawin Khametong, Kittipon Kankhunthod, and Chanon Warisarn

School of Integrated Innovative Technology (SIITec), King Mongkut's Institute of Technology Ladkrabang (KMITL), Bangkok 10520, Thailand

This work investigates a jointly optimized dual-reader shingled magnetic recording system designed for an areal density of approximately 2.0 Tb/in^2, utilizing a track pitch of 25.25 nm and a bit length of 12.75 nm. By employing an asymmetric secondary head offset of −2 nm and a simplified 3-layer depthwise-separable one-dimensional convolutional neural network, the proposed scheme achieves superior bit-error-rate performance while reducing the number of learnable parameters by over 60% compared to baseline models. Simulation results confirm that our proposed system provides robust detection and effective mitigation of track misregistration.

***Index Terms*—Dual-reader shingled magnetic recording, Depthwise-separable convolution, readhead optimization, track misregistration (TMR), low-complexity CNN.**

I. Introduction

The persistent demand for ultra-high areal density (AD) in digital storage has driven conventional magnetic recording toward its physical limit, widely known as the superparamagnetic limit [1]. To address this challenge, shingled magnetic recording (SMR) has been widely adopted by overlapping recording tracks, resulting in narrower tracks that substantially increase storage capacity [2]. However, this overlapping architecture introduces severe two-dimensional (2D) interference, including inter-track interference (ITI) and inter-symbol interference (ISI), as well as track misregistration.

While machine learning-based detectors, particularly convolutional neural networks (CNNs), have demonstrated exceptional capabilities in mitigating ITI [3], their immense computational complexity remains a critical barrier for real-time hardware implementation. Deep network architectures consume significant power and silicon area, posing challenges for application-specific integrated circuit (ASIC) integration. The dual-reader was also adopted to retrieve more information as proposed in [3]. This study addresses these complexity limitations by proposing a joint optimization scheme that combines optimal readhead positioning and network complexity reduction.

II. Proposed System

The investigated system operates in a dual-reader SMR environment over granular media. The arrangement of the dual readers directly retrieves more information about the interference, requiring a high-complexity subsequent neural network detector.

A. Dual-Readers Configuration

As illustrated in Fig. 1(a), in the ideal position (TMR = 0 nm), the primary head (H1) is positioned at the centerline of the target track to capture the main signal. Meanwhile, the secondary head (H2) is strategically placed at an asymmetric cross-track offset of −2 nm, as indicated by the lowest BER performance in Table I. This precise positioning is designed to effectively capture side-track information without being overwhelmed by adjacent noise. The system's resilience is further evaluated under a track misregistration (TMR) event, as shown in Fig. 1(b), where both heads deviate simultaneously toward the adjacent track, necessitating a robust detection mechanism.

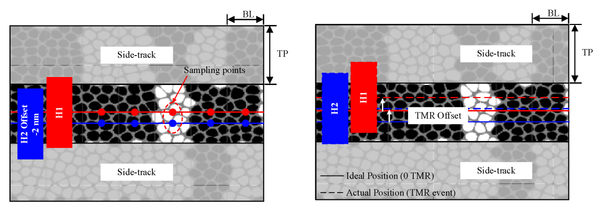

Fig. 1. Schematic of the head-media configuration in dual-reader SMR systems under: (a) ideal positioning (TMR = 0 nm) and (b) actual positioning during a track misregistration (TMR) event.

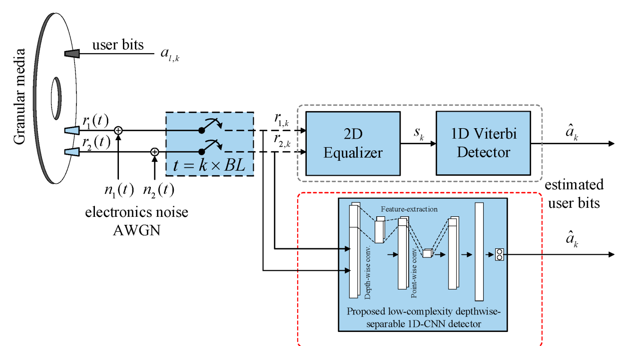

Fig. 2. Block diagram of the dual-reader SMR channel, with the proposed low-complexity depthwise-separable 1D-CNN detector (red dashed block).

TABLE I
Average BER Evaluation for The Secondary Readhead (H2) Offset Optimization

System Configuration	Readhead Offset	CNN Detector Architecture	Average BER
1) Baseline (Qin et al. [3])	12.63 nm (50% of TP)	13-Layer 1D-CNN	1.48×10^{-4}
2) Symmetric Displacement	3 nm	13-Layer 1D-CNN	2.50×10^{-4}
3) On-Track (No Offset)	0 nm	13-Layer 1D-CNN	1.25×10^{-4}
4) Proposed Optimal Offset	-2 nm	3-Layer 1D-CNN	3.85×10^{-5}

B. Low-Complexity Depthwise-Separable 1D-CNN Detector

The readback signal processing is depicted in the block diagram in Fig. 2. The user bits $a_{l,k}$ are recorded onto the media. During the readback process, the continuous-time signals from both readers $r_1(t)$ and $r_2(t)$ are corrupted by electronic noise, $n_1(t)$ and $n_2(t)$, which are modeled as additive white Gaussian noise (AWGN). The noisy signals are sampled to generate discrete-time readback sequences $r_{1,k}$ and $r_{2,k}$ from the respective readers. While the conventional system uses a 2D equalizer to produce an equalized signal s_k, for a 1D Viterbi detector, the proposed scheme directly feeds the raw sequences $r_{1,k}$ and $r_{2,k}$ into a low-complexity 1D-CNN detector to yield the estimated user bits $\hat{a}_k$. This refined architecture uses depthwise-separable convolutional layers to reduce the network from 13 layers (12 feature-extraction layers and 1 output layer), as in the baseline model [3], to a highly efficient 3-layer configuration (2 feature-extraction layers and 1 output layer) of our proposed scheme.

The discrete-time readback sequences are arranged as a $1{\times}1360{\times}2$ input data for the CNN. The training process employs 50 datasets generated under an SNR of 14 dB in the TMR-free condition. The output bit is estimated using a $1{\times}31{\times}2$ sliding window over the input data. The number of training epochs, mini-batch size, and learning rate are set to 15, 512, and 0.005, respectively.

III. Simulation Results

The performance of the proposed joint optimization is comprehensively evaluated through bit-error rate (BER) metrics under various signal-to-noise ratios (SNRs).

A. Readhead Offset Optimization

Table I summarizes the average BER performance of the asymmetric configuration, where the secondary readhead offset was optimized for the 1-, 3-, and 4-system configurations. By sweeping the H2 offset while keeping the primary readhead aligned on the track centerline, an asymmetric offset of −2 nm was identified as the optimal setting.

We also investigated the symmetric configuration by shifting both readheads equally with respect to the track centerline. The best average BER achieved under this condition was $2.50{\times}10^{-4}$ at a 3 nm offset. However, this result was still inferior to the optimal asymmetric configuration, which achieved a BER of $3.85{\times}10^{-5}$. This performance not only represents a substantial improvement but also outperforms the baseline 50% TP offset configuration ($1.48{\times}10^{-4}$) in [3]. The −2 nm offset provides an effective trade-off by capturing sufficient ITI characteristics for cancellation while avoiding excessive noise enhancement.

B. CNN Architecture Complexity Reduction

Following readhead offset optimization, Table II presents the architectural refinement of the CNN detector. The transition from a conventional 1D-CNN to a depthwise-separable CNN significantly reduces computational complexity while preserving detection capability.

TABLE II
Comprehensive Optimization of the 1D-CNN Architecture at the Optimal Offset (−2 nm)

Models	Conv. Type	Num. Layers	Pre-output	Params	BER
1) Conventional	Standard	13	Direct	16,502	$3.97{\times}10^{-4}$
2) Baseline [3]	Depthwise	13	Direct	3,167	$1.25{\times}10^{-4}$
3) Fully-connected (FC)	Depthwise	13	FC	3,167	$1.03{\times}10^{-4}$
4) Reduced	Depthwise	7	Direct	1,952	$5.80{\times}10^{-5}$
5) Proposed Scheme	Depthwise	3	Direct	1,142	$6.23{\times}10^{-5}$

Using the optimal readhead offset of −2 nm, several network architectures were evaluated. The proposed scheme employs only three layers, consisting of two feature-extraction layers and one output layer, while still achieving a highly competitive BER of $6.23{\times}10^{-5}$. In terms of model complexity, the proposed detector requires only 1,142 trainable parameters, corresponding to a 64% reduction compared with the baseline model in [3], which uses 3,167 parameters. It also represents a substantial decrease from the 16,502 parameters required by the conventional standard 1D-CNN.

The overall detection performance shown in Fig. 3 confirms that the proposed joint optimization maintains superior stability compared to conventional PRML-based detection and baseline systems. It remains robust against TMR-induced positioning errors, proving that a highly truncated network can perform exceptionally well when physical sensors are optimally positioned.

IV. Conclusion

We have presented a jointly optimized dual-reader SMR detection scheme. By combining an optimal −2 nm secondary head offset with a low-complexity depthwise-separable 1D-CNN detector, the system achieves a significant reduction in complexity without sacrificing accuracy. This synergistic approach provides a robust, power-efficient, and practical framework for enhancing the reliability of next-generation high-density magnetic storage.

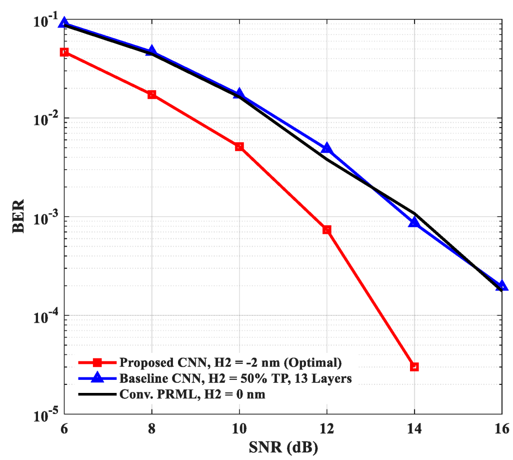

Fig. 3. Comparison of bit error rate (BER) performance versus signal-to-noise ratio (SNR) for the conventional PRML, baseline CNN, and proposed optimized CNN architectures.

Acknowledgments

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